Comparative Q-learning and Heuristic Chemotaxis Agents For Simulated Microscopic Movement

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Chemotaxis is a means of blind navigation and transportation for microscopic organisms through a complex microscopic world. The process effectively guides organisms into positive chemical gradients and away from negative chemical gradients without the complex structures required to visualize a space. Navigation by way of chemotaxis occurs based on chemical signals and experience associated with the history of such signals. There is an underlying mathematical approach that chemotactic organisms take to successfully perform chemotaxis, using chemical signals to identify trends and optimize for future success. It is this mathematical angle in which chemotaxis can be identified, which opens up the opportunity of chemotaxis simulation through mathematically sound and logical Reinforcement Learning (RL) processes. Two RL approaches are employed throughout this paper to attempt a chemotactic-like optimization of an agent into a gradient of helpful chemicals: a Q-Learning model and a heuristic-based model. These two models directly reflect the behaviors of a microorganism going about chemotaxis by simulating the pattern recognition and learned experience required for successful identification of a positive gradient field. The ability of such models to optimize chemotactic behavior is defined by hyperparameters of alpha, gamma, and epsilon, with the ability to influence learning, and thus influence the ability to recognize and move based on trends. It is the difference in general behavior and behavior in reaction to certain hyper-parameters by the models involved in this paper that sets the bounds for successful chemotaxis simulation for each of the respective models. From this comes a comparison of model success and model influences, which can effectively define what models optimize most realistically and can be best applied to future microscopic innovations, including microscopic robots and all other interaction of humans with the microscopic world.

Keyword 1 $Chemotaxis$ - Keyword 2 $Q-Learning$ - Keyword 3 $Heuristic$ - Keyword 4 $Optimization$ - Keyword 5 $Steepness$ - Keyword 6 $Hyperparameter$

To consider the utilization of RL strategies to simulate the processes of Chemotaxis, it is necessary to understand the characteristics of both topics individually before exploring the ways in which such complex biological processes can be simulated effectively by modern computing.

A. Chemotaxis. The process of chemotaxis regulates the motion of microscopic organisms, serving as a sensory mechanism that provides guidance through various obstacles and features of microscopic worlds. Chemotaxis aims to achieve the general goal of survival, including, as explored in this very paper, the identification and tracking of beneficial chemical concentrations.

An organism must make chemical analysis and informed evaluation of a chemical gradient in order to navigate. The ability to pick up on chemicals comes from chemical sensors, allowing chemicals to be differentiated based upon positive and negative indicators. These chemical signals are identified for the current state of an organism, meaning the concentration of beneficial and harmful chemicals identified is related to a specific experience ([1]). Tracking a superficial history of chemical signals alongside a trend in movement from location to location allows an organism to identify basic trends for successful movement into gradient fields.

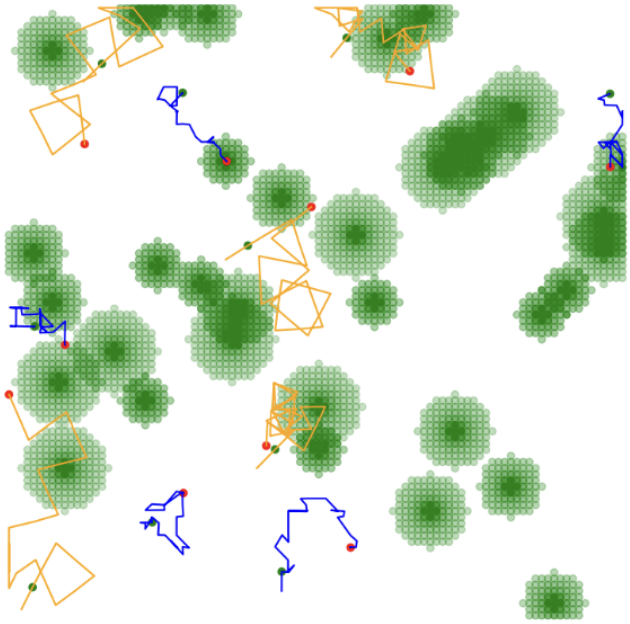

Fig. 1. Traversing gradient fields of positive chemicals in a large simulation space (Blue: $QLearning$, Orange: $Heuristic$) (Expand on this for the final paper)

The ability to interact with gradient fields is based on the decrease in the chemical concentration with increased distance from the gradient center. This consistent definition of gradients allows an organism to reliably track concentrations and climb

Significance Statement

Chemotaxis is a complex biological process which has been optimized over millions of years to effectively drive the motion of microscopic organisms traversing through chemical gradients. The simulation of such a complicated process enables understanding into how organisms interact with microscopic worlds, making it possible to bridge such interactions into the interactions of humans. As knowledge of processes such as chemotaxis become more extensive, it will be possible to provide life saving guidance to microscopic robots and manmade agents, following the guidelines of the most well proven microscopic path finders: chemotaxis driven organisms.

All code and simulated models applied in Python on the Google Colab Jupyter Notebook compiler.

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in the direction of increasing concentration. Organisms utilize comparison of chemical readings with movement through a gradient field, relating back to chemical signals as well as considering the current state and chemical signal in order to climb into positive gradients, following the path of increased concentration.

The reactive process which is performed at the heart of chemotaxis navigation is often considered in a *run and tumble* scenario, in which an organism has the ability to react in a run or a tumble ([1]). A run action performed by an organisms refers to the movement in a specific vector direction, and tumble defines the possible step to step change in such direction. Traversing through the chemically filled space at a specific vector, an organism can use recent chemical concentrations during its path in order to determine if a tumble action is necessary. A tumble action by an organism is in reaction to navigation failure, in which the vector taken by an organism fails to bring about ideal chemicals, resulting in a necessity to redirect motion and seek success in other directions. The basis of the two chemotaxis models employed in this paper is in such run and tumble behavior, relying upon reactive vectors for subsequent moves in the action space.

B. Reinforcement Learning. Reinforcement learning (RL) is based on the optimization of policy through experience gained in agent interaction with an action space, including states, actions, and defined rewards. Agents interact with the action space, informing the perceived values of actions from one state to another, and updating model policy for success [2, 3, 4]. In this paper, reinforcement learning in Q-Learning and rule-based heuristic models are outlined for the sake of comparison, both defined similar to other research in this field, as smart artificial agents performing simulated chemotaxis through RL [5].

Q-learning in this paper context is employed as a form of RL with a policy based in the maximizing reward defined by the collection of values within a Q-table, each Q-table value representing the perceived benefit of being within a given state and transitioning out with an action into a new state. The Q-learning policy is based in the comparison of such Q-table values, optimizing to maximize cumulative reward of a simulation period. Q-learning as used in this paper is a model-free approach, a form of RL in which the Q-table values are updated based on the simulated experience. A model-free Q-learning model begins with an effectively empty Q-table in which all values are empty for all possible state-action pairings. The system purely gains knowledge from training episodes, in which an agent simulates episodes of chemotaxis in order to update the Q-table for state-action pairs. This simulation process brings about convergent reward values that can confidently drive later simulation.

The heuristic model is based upon a series of unique guidelines provided prior to runtime, allowing complete control over the model, and covering possible states of the system with actions based on an optimal rule-based policy. Such direct control over the actions/reactions of an agent functions exclusively with an understood situation of chemotaxis, as outside of such cases it is likely for rules to lack the necessary scope of a scenario, limiting the possibility of true optimization, and possibly failing as a system. With reasonable understanding of the gradient field defined in the uniquely developed heuristic model policy, there is confidence in model performance similar to that of the Q-learning model, marking the possibility for later comparison in this paper.

Sections: Methods, Results, Result Synthesis, Conclusions

In the following sections of this paper are the methodology of models, the results of the simulation, and the conclusions of the results. All areas aid in the analysis required to approach the task of comparison for models aimed at effective and usable chemotaxis replication.

Consider the following essential variables which are present in various equations throughout the following sections of this paper. These variables are clarified and described to a greater extent in specific cases later on.

For Gradient and Models:
d = *distance to center at state*
s = *Steepness*
r_p = *reward at current position*
p = *position (x,y)*

For Heuristic:
θ = *direction of movement in degrees*
v = *velocity of movement*
m = *constant for model adjustment*
σ = *progression of perceived reward*

For Q-Table:
S_t = *current state at step t (x,y)*
A_t = *action taken at step t*
R_{t+1} = *reward of state reached by* A_t
α = *learning rate*
γ = *discount factor*
$max_a Q(S_{t+1}, a)$ = *max future value for* S_{t+1}

1. Methods

Factored into the success of chemotaxis simulations is the direct relationship between reinforcement learning strategies and the biological processes. With deep analysis into both chemotaxis and reinforcement learning methods, it was possible to craft intuitive RL policies that can drive simulated chemotaxis by optimizing motion between subsequent positions in a larger gradient field [6, 7, 8].

A. Gradient Field. The basic ideas of chemotaxis lie in the position within a gradient field, in which there are centers of chemical concentration that fade into the surroundings. This gradient field must be defined around an equation of decreasing chemical concentration with an inverse relationship between concentration and distance from chemical center. For the sake of this paper, agents will be simulated within a square grid, understood to be 20x20 units by the agent, with a central chemical gradient field of positive chemicals. The relationship of decreasing chemical concentration from the central gradient outwards is limited to only regions within 5 units of the center, as chemicals are cut off outside of the region.

In order to relate the nature of the central chemical gradient to the perceived reward of the digital agent moving into a specific state within the action space, it is essential to define a reward equation that establishes a connection between position

and detected value. The specific reward of a state in the established gradient field is partially defined by a variable for gradient steepness, with steepness setting the rate at which chemical concentration decreases from the center. Gradient steepness will serve as a point of model comparison later in this paper, and is set to 0.75 for all model optimization as a balance between overly steep gradients which prove unclimbable, and overly shallow gradients which are simply unrealistic.

The function for reward at a given position is defined under the following expressions and conditions related to agent positional information and predetermined gradient steepness:

$$r_p = \begin{cases} \frac{10}{\mathrm{d}^{\mathrm{s}}}, & 1<\mathrm{d}<5 \\ 50, & \mathrm{d}<1 \\ \text{-}50 & \mathrm{d}>5 \end{cases}$$

B. Brownian Motion. Brownian motion during provides a means of random variation from step to step of the simulation process [2]. This random variation in the motion of the agent reflects the reality of microscopic particles with trajectories impacted by random particle interactions with small alterations in direction. In this paper, Brownian motion randomly offsets the position of the agent within the range of -0.1 to 0.35 units in the simulation space. The focus on a random but by majority positive output for random variation in both the x- and y-axis for agent position relates to the predetermined trend of particles to assist the agent in early navigation, allowing the establishment of trends towards the centralized chemical gradient.

C. Approaches. With an well outlined understanding for the gradient field action space, it became essential to define the reinforcement learning approaches to actual chemotaxis replication. This paper approaches such replication with an essential question: what approaches can theoretically lead a digital agent through a gradient space in a way that mirrors real chemotaxis and the properties of the process? The strategies closely analyzed by this paper and well explained in the following paragraphs are model-free Q-learning and a rule-based heuristic systems. Although the general logical approach of these two systems are distinct in nature, it is notable that the shared goal in the reflection of true chemotaxis provides the provision for this papers model comparison structure.

C.1. Q-Learning. The model-free Q-learning approach to chemotaxis is defined by a balancing act between optimization of Q-table values and random exploration from an adaptive epsilon, attempting to find a policy that can exploit both to closely resemble chemotaxis. The issue between the extremes at each of these is that greedily defined optimization does not effectively simulate true chemotaxis as it is too perfect, and random exploration will never effectively pick up on patterns, making chemotaxis effectively impossible.

Structuring the Q-table for later Q-learning model optimization was accomplished with entries defined as an initial position (state) and move in direction and velocity (action) which result in a reward and a next position, formatted as $[(initialX, initialY), (direction, velocity)]$ paired to $[calculated - reward, [nextPosX, nextPosY]]$. This format simplifies to a state-action pair related to a Q-table value and resultant state, mirroring the basic Q-table structure required for all Q-learning optimization [9]. Actions formatted as direction and velocity are limited to a velocity between 1 and 4 units per simulation step, and a direction 0 to 360 in steps of 45 degrees. The range provided in velocity for step to step actions for the chemotactic agents provides an opportunity for adaptive logic in connection with gradient concentration, allowing the agent the ability to adjust to slower velocities upon high reward signals [10].

In order for the model to optimize, an agent will run through a series of 300-500 training episodes, in which the model begins with a random first move from (0,0), and then uses the most up-to-date Q-table entries and reward values to either maximize the reward, or move randomly from the current state over 50 training steps – random motion occurs at a predetermined frequency of epsilon. With each step of a training episode, the value for a state-action pair in the Q-table is updated through the Q-table update equation outlined below, taking into account the current Q-table value as well as possible future rewards in relation to a specific state-action pairing followed by the agent. The Q-table update effectively provides the model a means of learned experience, which through continuous contact of state-action pairs leads to a convergence of Q-table values [9].

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t) \right]$$

With a well trained Q-table it is expected that an agent can act greedily with its surroundings, having the necessary means in the form of convergent Q-table values to identify the optimal action from a given state, moving through the action space and up the chemical gradient efficiently. However, this greedy movement as a policy is not reflective of chemotaxis, an issue addressable through an adaptive epsilon value. The adaptive epsilon adjusts based upon the agents location, aligning with the requirement of increased randomness due to a lack of gradient information outside of the central chemical field. If the agents exists outside the gradient, epsilon will be increased from the standard 0.15 to 0.4 to drive more optimal tendencies in the policy. The specific increase of the adaptive epsilon value was determined by analysis over agent performance in simulated pathways with differing pairs of standard and adaptive epsilon values(*Figure 2*). There was a necessity to balance epsilon and adaptive epsilon to ensure reasonable use of exploration and exploitation of the model to simulate both chemotactic randomness (exploration) and gradient climbing (exploitation).

For training and simulation of the model, the qualities of performance and optimization are tested throughout this paper in response to alteration in alpha and gamma values used within the Q-table update equation. As hyperparameters to the Q-table update equation, alpha and gamma share the ability to change the equations behavior and redefine the ability of the system to process reward information throughout the action space and to relate such information to state-action pairs of the Q-table [9]. The impact of alpha and gamma is explored to a greater extent in later sections of this paper, analyzing the tangible impacts on chemotaxis agent behavior resulting from a Q-table impacted by alterations in the hyperparameters.

C.2. Heuristic. Development of the rule-based heuristic model for chemotaxis simulation required a strict focus upon subsequent moves in agent interaction within the gradient space, utilizing tracked gradient information in order to drive opti-

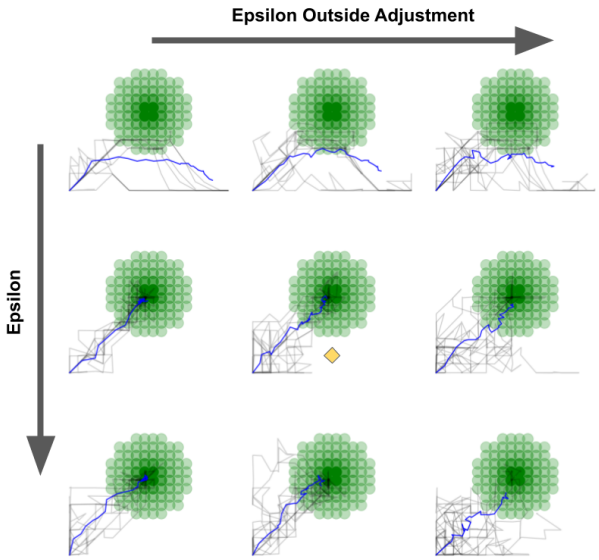

Fig. 2. Q-learning agent performance in simulated navigation pathways at differing standard and adaptive epsilon values. Average pathway shown in blue as well as 10 samples of pathways in black, layered on one another. Yellow diamond at the center of the figure indicates samples for agent behavior produced by the model defined with the values of epsilon and the epsilon adjustment utilized in the actual experiments of this paper.

mal movement in the heuristic policy. Information is taken from the environment in the form of concentration readings, which are processed as reward values. Reward reading serve as knowledge fo the agent, helping to update a running value of perceived reward, with the comparison of this perceived reward prior to and following a simulation step defining the policy for movement and thus all agent behavior.

The perceived reward is calculated through a unique equation that is based on a similar design to the Q-learning update equation, such that the past reward is changed to the current reward following the addition of the update value with hyperparameters at work. This equation is defined below with m representing a multiplier for the sake of model tuning, with iterative testing for values of m defining it for the final model:

$$\sigma_f \leftarrow \sigma_i + m * \alpha \left[r_p + m * \gamma * r_p - \sigma_i \right], \; m = 3.75$$

The update to the perceived reward value considers the change from the past reward as well as the perceived reward of the next position, providing the agent multiple layers of interpretation between agent movements just as with the Q-learning approach. Required variables included in the update equation are adjustable values of alpha, and gamma values. Working alongside one another, these two hyperparameters define how much the information at a respective position may be weighed as the model runs. It is the alpha value which directly decides the weighing of the new information as a whole, while the gamma value more specifically decides the weighing of the possible of rewards in the future from a certain position.

The rules rely directly on the running and updating value of perceived reward σ, with such rules directly defining the direction and velocity of the system. Both velocity and direction are recalculated after each step according to the model policy, evaluating perceived rewards and the current position.

The velocity rules within the policy do not require a comparison factor of the perceived reward to lead to a final result. Velocity is defined by the current position of the agent, with a better position leading to a lower velocity and a poor position leading to a higher velocity. This logic revolves around the idea that the agent should prioritize high reward, not moving out of such areas with high velocity, and avoid areas of low or negative reward, quickly moving away with high velocity [10]. This equation for velocity is defined around current reward r_p:

$$v = \begin{cases} 4, & r_p < 0 \\ \frac{1.5}{r_p}, & r_p > 0 \end{cases} \qquad 1 < v < 4$$

The direction rules of the policy rely on comparison and relative positioning to decide on the new direction that the agent should take. The rules which define direction throughout model simulation begin with an analysis of relative position, isolating whether or not the agent is located within or outside the gradient with different equations of direction change for following steps applied over the model.

When the agent is located within the gradient and must define a new direction for the next move, the change in direction is defined as such:

$$\Delta\theta \leftarrow \pm 360 \left(\frac{\sigma_i}{\sigma_f} \right)^m, \quad m = 0.8$$

This equation is backed by the fact that with the agents position within the gradient, the current perceived reward value will always be greater than the past, making $\frac{\sigma_i}{\sigma_f}$ a value less than 1. Logically, if the final perceived reward is increasingly larger than the initial perceived reward, the change in direction in degrees will be increasingly small, leading the agent in a similar direction, favoring prior success. If the current and past perceived reward values are incredibly close to one another, the change in direction will be relatively large, favoring a change in the agents directional strategy after less promising results.

When the agent is located outside the gradient and begins to process to making a new move, the change in direction equation for the sake of the upcoming move is defined by a mathematically optimized version of the following expression:

$$\Delta\theta \leftarrow 180 \pm 180 \left(\frac{min(|\sigma_i|, |\sigma_f|)}{max(|\sigma_i|, |\sigma_f|)} \right)^m, \quad m = 0.6$$

Before discussing the logic behind this equation that allows it to relate with chemotaxis simulation, consider the use of minimum and maximum over the values of the absolute value of σ_{past} and $\sigma_{current}$. This expression ensures that out of the two values, the absolute smallest is placed in the numerator and the larger is placed in the denominator, regardless of sign being positive or negative. This structure ensures that a value smaller than 1 is multiplied over the term ± 180, backing up the logic of the entire portion of heuristic policy.

Assuming that the basic equation and the alteration to it run correctly while interacting with the gradient field throughout simulation, there is logical backing behind the rule as a whole. Since it is guaranteed that the numerator of the σ related term of the expression is smaller than the denominator term, the variation between σ_{past} and $\sigma_{current}$ defines the policy behavior. The smaller the variation of σ values, the

smaller the value returned by the equation of change in direction for the agent, allowing the agent to prioritize directions that avoid detrimental impacts on perceived reward. The greater the variation of σ values, the greater the change of direction returned in the direction change expression of the policy, aligning with the idea that large changes in perceived reward outside the gradient relates to entrance into the negative regions outside the chemical center, meaning ideal motion is reversing direction.

2. Results

With well defined models and means of simulation, it became possible to make continuous runs of the chemotaxis simulators, making alterations in various qualities of both the model and the field in which the agent is meant to traverse. Data collection for each chemotaxis approach across such variations provided the means for analysis through comparison of agent behavioral reactions, defining the impact of this research work in connection with chemotaxis replication.

A. Alpha and Gamma Analysis. Alpha and gamma are key hyperparameters of both the Q-learning and Heuristic models of simulation, with testing for alterations in hyperparamters making it possible to pull key behavioral trends of both approaches. Analysis over graphs of average distance to the center at final step, average cumulative reward, average number of steps inside gradient, and average number of failures during model simulation for both approaches at various alpha gamma pairs helped to paint the picture of model nuances and features.

A.1. Q-Learning. Analyzing the results of numerous iterations of Q-learning chemotaxis model simulation, pictured with detail in *Figure 3*, it was possible to determine the behavioral aspects of the Q-learning model. Testing across a series of alpha-gamma pairs exposes the nuances of model in reaction to changing information and learning rates, with pairs of hyperparameter covering cases of alpha and gamma between 0-1.

The general trend seen between success in metrics and alpha-gamma pairs is higher alpha and gamma values leading to a greater number of steps in the center gradient but a less clear benefit with other metrics. While in the other measures of agent performance, there is more success in alpha-gamma pairs that are not near the absolute minimum, a relationship does not seem to drastically connect the metrics beyond that. The differences in trends between performance metrics suggests nuances to Q-learning optimization and simulation, breaking the expectation of the Q-learning model to fully benefit with increased gradient information from hyperparameter increase.

Defining the significance of the alpha and gamma pairs as hyperparameters controlling the rate of learning for a model during training, high values in hyperparameter pairings relate with extensive gradient information while low values in hyperparameter pairings relate with limited gradient information for the Q-learning agent [9]. With this it is possible to assert the importance of accessible gradient information for the performance metrics given. With a clear relationship between hyperparameters to success, gradient information is essential in the average steps the agent takes within the gradient. On the other hand, there is not a significant relationship of the

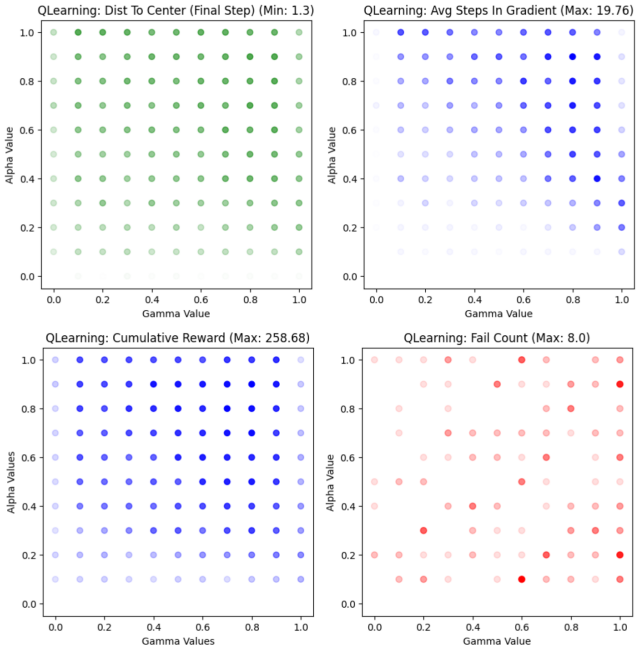

Fig. 3. Comparison across $alpha - gamma$ pairs for distance at final step, number of steps in gradient, cumulative reward, and number of failures.

Q-learning performance in cumulative reward, average final distance to the center, and number of failures of the model, to alpha-gamma pairings, suggesting the special ability of the Q-learning agent to bring about optimal navigation despite limitations in gradient information.

From this lack of reliance in some cases but a necessity upon gradient information in other key measures of the Q-learning success, there is a duality to the Q-learning representation of chemotaxis. The model maintains the adaptability and versatility which comes from not requiring a serious amount of gradient information, while also representing the trend of increasing information carrying the provisions for heightened success. This type of duality in the Q-learning model is essential in driving model success in a multifaceted sense.

A.2. Heuristic. Approaching equivalent samples of results for the sake of the heuristic model of chemotaxis, notable trends in behavior and properties are exposed in the heuristic model's reaction to a set of alpha-gamma pairs including the parameters in range 0-1 in all combinations.

A clear trend exists for the heuristic model and various alpha-gamma pairs which can be picked out from analysis over data collected across hyperparameter variation, seen in *Figure 4*. As a general trend, the behavior of the heuristic agent is the most optimal for the given metrics within a certain set of alpha gamma pairs, including most generally the higher alpha and gamma values (0.5 and above), and more specifically including a specific band of high hyperparameter values.

The general trend reflects expectations of reinforcement learning models in relation with alpha and gamma values, as high values of the hyperparameters represent a models heightened ability to read its state, in this case the gradient field, and react well to changes in the readings. However, the specific region of alpha and gamma pairs at non entirely maximized alpha and gamma levels suggests unique aspects

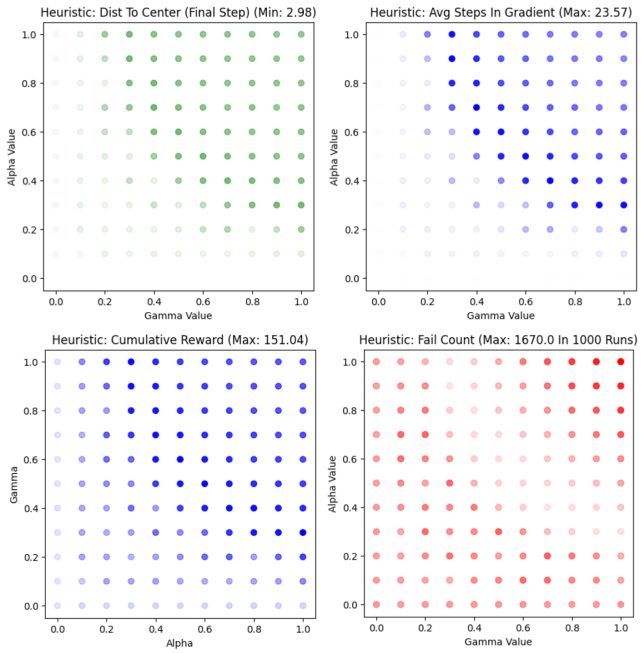

Fig. 4. Comparison across $alpha - gamma$ pairs for distance at final step, number of steps in gradient, cumulative reward, and number of failures.

of the heuristic model, as the best results in all collected performance metrics are clearly brought about by either non maximized alpha and gamma or a more maximized value for a single parameter and a less optimal value for the other.

Nuance in the metrics of optimal navigation for the heuristic model illustrates the reliance upon hyper specific alpha and gamma values for optimal navigation into the chemical gradient during simulation of the heuristic agent. This reliance suggests the heuristic models direct requirement of specific readings from the gradient field to function well.

The model has the incredible capability to optimize with maximum effectiveness when provided specifically trackable gradient information, aligning with the suspected reaction of an agent to ideal inputs of reward information. However, the model will fail to optimize to the same level with any deviation from such specific gradient information, taking away a level of adaptability and versatility which can prove necessary in variable gradient conditions.

B. Optimal Results.

B.1. Q-Learning. With an established base in the impact of hyperparameters onto Q-learning model performance, it became necessary to discuss the features of the Q-learning model when trained and simulated for numerous episodes at optimal hyperparameter conditions, using information in *Figure 5* to analyze the ability of the Q-learning model to perform successful and realistic chemotaxis. These optimal conditions were determined in the prior section of results under the analysis of alpha-gamma pairs and the resulting behaviors of the agent. Utilizing metrics from optimal conditions makes it possible to measure the ceiling of the model, determining what works and what fails at best case in bringing about realistic chemotaxis.

Analysis of the average distance to the center at all steps in the simulation process depicts a gradual approach to the gradient center, becoming incredibly close around halfway through the steps. This drawn-out approach into the chemical field on average is related to a random exploration of the first steps and a gradual, but effective, climb of the gradient following initial contact. Furthering the responsive nature of the Q-learning model at optimal conditions was the clear ability for the agent to, on average, never head away from the gradient center, instead gradually approaching and then solidifying position towards the center of the gradient, maximizing reward and proving its effectiveness.

As a result of the averaging done for the distance from the center at all steps in the simulation process of the Q-learning model, key information about the behavior in specific cases of optimal conditions is lost. Thus, there was serious importance in evaluating for the standard deviation bars and the samples of simulations of *Figure 5*.

The standard deviation bars illustrate the consistent variation of the agent under optimal conditions for distance to the center at all simulation steps, resulting from the fundamental randomness of the model. The sample of 50 simulations of the Q-learning model furthers this understanding of variation, making it possible to point out the consistency of the Q-learning model to approach the center of the positive gradient, and remain within the central gradient despite variation. Together, the variation in the simulations of the Q-learning model and the consistency in the model reaching the inner chemical field solidifies the ability of the Q-learning agent to perform successfully at optimal conditions.

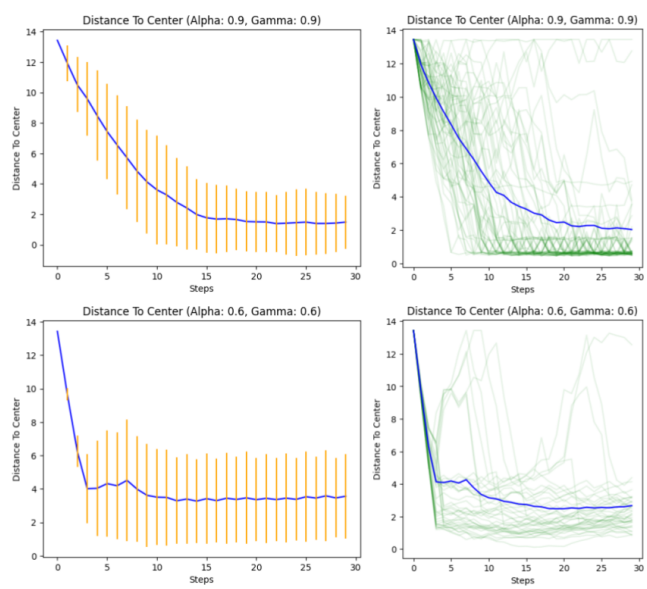

Fig. 5. Standard deviation and sampled simulation graphs for *Q-learning* (upper graphs) and *Heuristic* (lower graphs) models.

B.2. Heuristic. The analysis of Heuristic model performance at experimentally determined optimal conditions, meaning optimal in reward and trajectories, was highlighted as the measure of the maximum ability of the heuristic approach to chemotaxis in agent behavior. It is only under the analysis of optimal performance in *Figure 5* that the unique qualities of the heuristic model under optimal conditions can be determined and set for comparison against the expectations of true

chemotaxis, thus identifying the features of the model that provide success and the qualities indicative of model failure.

In considering first the general features of the graph of average distance to center, there is a notable drop in the distance to center, followed by a period of stabilization in which average distance sees minor or completely absent changes. The early drop in the distance to the center connects with the heuristic agent's guidance at early steps in simulation, as the model clearly draws the agent directly into the gradient in only a few steps on average, lacking truly exploratory qualities expected out of a *run and tumble* agent [11]. This is an undeniable limitation of the heuristic model, as having such a lack of exploration throughout the action space early on in simulation signals an over-optimization of the model, which robs the agent of more random motion required for uninformed chemotaxis.

The understanding that was gained from the metric of average distance from the center was deepened by analysis into the standard deviation bars, and into a series of sample simulations for the heuristic model. These metrics expanded upon the details hidden behind the averages of distance to center. Considering the standard deviation bars at optimal conditions, there is a clear lack of deviation in early steps, but serious deviation once the model reaches the central gradient field. Such behavior adds support the nature of the heuristic model to consistently drives the agent towards the center gradient immediately, without any true variation, and then settles inside the central gradient, where more randomness at small step sizes still represents optimal motion.

The series of sample simulations of the heuristic model solidifies the consistency in heuristic model over-optimization, while also bringing up a fascinating ability of the model. Despite the excessive optimization early on, the heuristic approach proved a unique capability of rerouting once exiting the gradient field, as there are visible cases in which the agent ends up far from the center and quickly returns back into the gradient field by reversing the prior negative movement as seen in *Figure 5*. Although this type of extreme corrective optimization in a gradient fails to relate to chemotaxis, the unique ability to reroute the agent following navigation failures seemingly connects to an increased ability of the agent to react to changes in gradient information, presenting agent behavior resilient in nature with protection against agent failure.

C. Gradient Steepness Analysis. Exploring the bar graph data of *Figure 6* exposes the responses of Q-learning and heuristic models to changing gradient steepness. Gradient steepness is varied from 0.1 to 1.9, meaning the two chemotaxis models are tested for behavior within gradient fields of high overall and low overall concentrations heading outside of the chemical center, exposing the level of reliance upon sufficient reward values.

The heuristic model reacts poorly to increasing steepness of the chemical gradient, as the chemotaxis simulation agent struggles to establish the necessary relationships in order to run correctly. Resulting from increased gradient steepness is a decrease in gradient information, meaning chemical signals in the form of rewards are highly limited during simulation. Such a lack of gradient information proposes an impossibility of success for the heuristic system as the model is rooted in sufficient gradient information in order to support optimal behavior – weakening the case of the heuristic model as a realistic simulator of navigation.

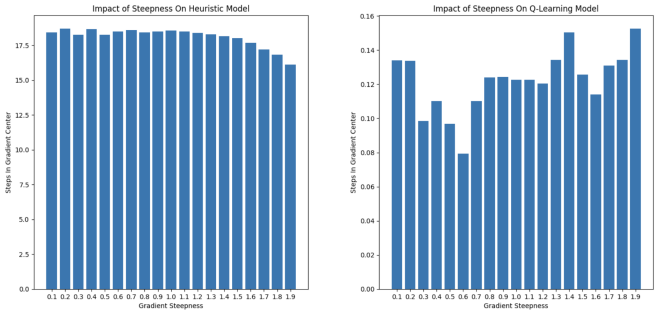

Fig. 6. Impact of steepness upon chemotaxis models steps in central gradient

The Q-learning models response to changing gradient steepness fails to establish a general trend in the relationship between positive agent performance and a decrease or increase in gradient steepness. The lack of impact gradient steepness has over the Q-learning model performance connects with the Q-learning abilities as a reinforcement learning technique to rely on relationships between rewards as opposed to the true value of individual rewards. Such a reliance purely upon reward relationship allows the Q-learning model to pull information around the gradient field regardless of steepness, establishing to the Q-learning agent the gradient trend of increasing rewards heading towards the center. This related reward information is pivotal to gradient climbing which was performed with resiliency in the face of changing or lacking gradient/reward information

3. Result Synthesis

A. Q-Learning. The abilities of the Q-learning approach to chemotaxis, as well as any other kind of update table based reinforcement learning, comes down to the root of RL approach, that being the Q-table and the update equation associated with it. The optimal behavior observed by the Q-learning agents in various key simulation metrics in the results section of this paper have been reviewed to have come from the basis of Q-learning as a reinforcement modeling approach, making it imperative to solidify what is at the bottom of Q-learning success and in some capacity what explains some failures.

Successes seen in the Q-learning simulation of chemotactic motion in part relate directly to the heightened ability of the Q-learning model to consistently simulate despite changes in essential gradient information, either by way of changing gradient steepness or learning hyperparameter. The non reliance comes directly from the fact that the Q-learning approach does not rely upon specific gradient information, instead relying upon the relationship between gradient values at different points. Relationships can be established regardless of specific reward information as the basic gradient relationships of increasing gradient value heading towards the center will always be understood. This consistency in the Q-learning model to navigate based upon learned relations in the gradient field is by far the greatest strength of the Q-learning model, relating directly to the abilities of chemotactic organisms to track chemical information with great consistency, establishing relationships based upon such information in order to track optimally into positive chemicals.

The ability of the Q-learning model to effectively track

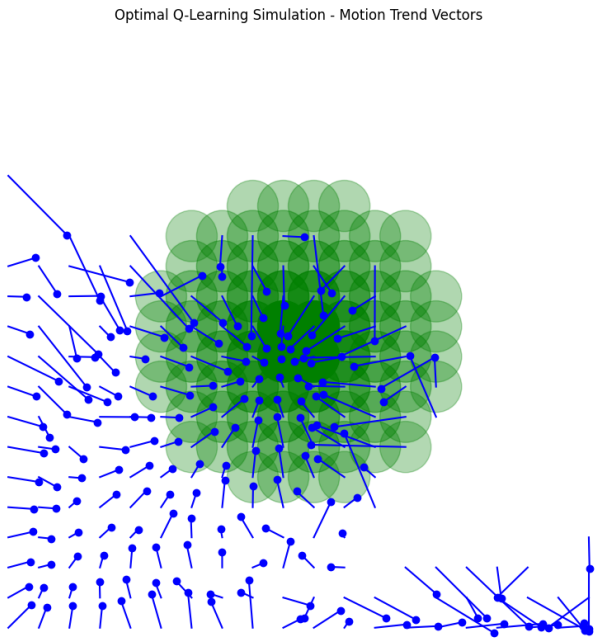

Fig. 7. Vector field of the average vector taken from states contacted by the agent across 1000 Q-learning simulations, points representing the end position resulting from vectors taken.

into the gradient center, as exemplified throughout the results section of this paper, can be explained best in the summarizing of Q-learning abilities through the plot for an average vector field across numerous simulations (*Figure 7*). It is clear through such a vector field that there is significance bias in the direction of gradient center, as all positions interacted with by the agent point generally towards the higher gradient concentration. This is the expected behavior of an optimal model, effectively ensuring that the average agent will orient and traverse through the action space in the direction of higher rewards. The failure of the vector field to reflect this bias in the direction for gradient center in the bottom right corner of the action space reflects not a model failure, but rather a nuance of model optimization and training. As the model becomes more optimized in Q-table update logic, it is less likely for the agent to interact with the states far from the optimal path, such states including those in the bottom right of the action space. As a result of less interaction with the states as the model optimizes, there is limited Q-value knowledge over the states, resulting in vectors which appear uninformed but are simply unnecessary in an optimal context.

The random motion built into the Q-learning model adds another layer of chemotaxis reliability. Simple descriptions of chemotaxis depict a *run and tumble* style of motion, in which an organism will move, or run, through the field around it searching for positive chemical information, changing direction, or tumbling, when such information is not found or when negative chemical information is identified. For the case of the Q-learning model, such *run and tumble* behavior is seen directly in early stages of simulation in which the model moves with variation when not within the central gradient with chemical information, as the model lacks the chemical information required to end the tumbling stages and climb the gradient [11]. If the Q-learning model were not to act with a level of randomness early on into the simulation of navigation, the model would fail to reflect what occurs in chemotactic organisms in nature, as lacking gradient information should spur varied motion as seen in results not over-optimization which tracks directly into the central gradient.

Sprouting from the consistency of the Q-learning model in tracking gradient information after the gradient is contacted following a series of random actions early on in the simulation process, is a serious weakness of the Q-learning model as in order to perform in such an optimal manner it is essential for the Q-table, that drives simulation, to have significant enough gradient information to establish relationships that may lead to navigation. The reliance upon reward information rooted in a table prior to the beginning of the Q-learning models simulation, as seen in all results, rejects some of the qualities of chemotaxis simulation, as prior knowledge turns against the foundation of the motion being a detection and reaction based navigation strategy. Despite this weakness in the Q-learning approach to chemotaxis there are redeeming qualities in the requirement of gradient and reward information prior to simulation begins. Real chemotaxis requires countless biological foundations in order to effectively go about microscopic gradient fields and identify gradient information, making it possible to argue the equivalence of such biological factors to the organism having prior gradient understanding. While such equivalence may not seem obvious at first, it is essential to consider how biological factors provide the organisms with a level of prior knowledge over how to generally navigate into gradient fields, making the Q-table information a simple reflection of having a level of general action space understanding.

B. Heuristic. The abilities of the heuristic approach to effectively simulate realistic and efficient chemotaxis can be best understood by way of the logic which backs up the computerized navigation strategy. The logical basis which defines the heuristic model of chemotaxis is directly reflected in the results of model optimization outlined in the prior section of this paper, as simulation performances that relate to chemotaxis are directly explained by inherent heuristic features.

As an approach to chemotaxis simulation, the heuristic model is based upon two key logical functions for velocity and direction, directly controlling the rate at which the agent moves in a gradient field as well as the direction in which the agent takes moving at a given speed. The function for agent direction between simulation steps is directly reliant upon the progression of perceived rewards during simulation, meaning the chemical readings in the form of reward values identified by the agent during movement are fully linked to gradient information and the ability to read gradient information. This understanding of the process of direction change between steps of heuristic simulation illustrates a key idea of the approach in which the model is highly reliant upon the establishment of specific relationships in the perceived rewards in order to navigate in the correct directions at the correct rate.

The reliance upon specific reward information in the heuristic approach to chemotaxis simulation is proven out by all significant metrics of agent optimization explored throughout the results section of this paper, as it is clearly showcased the inabilities of the heuristic model to perform outside of optimal conditions. Optimal conditions are offset by changes in gradient steepness, and the hyperparameter of alpha and gamma,

with even minor changes in such values leading to significant change in model simulation results away from success. This speaks to a serious complication with the heuristic approach to the biological process, as there is a simple inability to adapt with changes in the gradient space in agent interactions. In terms of a real microorganism, a small creature without the capabilities to adapt and perform optimally even in less than optimal or more variable conditions will fail to pick up on key chemicals, thus failing navigation outside of perfect conditions and likely failing to survive for long period of time.

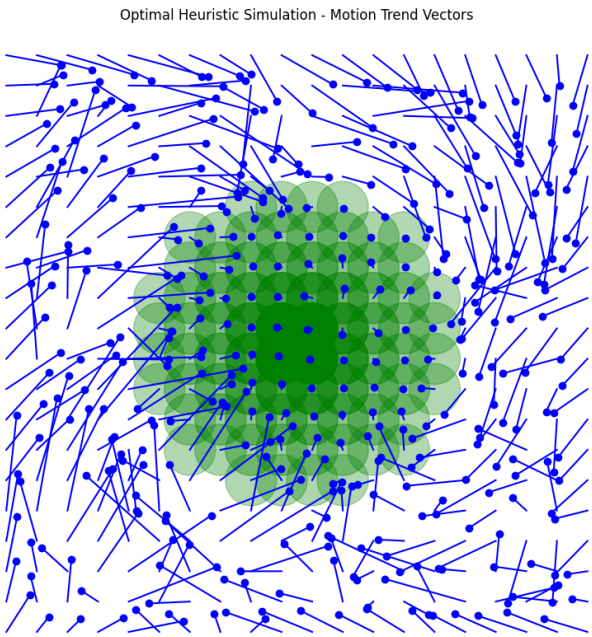

Fig. 8. Vector field of the average vector taken from states contacted by the agent across 1000 Heuristic simulations, points representing the end position resulting from vectors taken.

While the issue of unreliability and inability to handle variability strongly opposes the use of a heuristic approach to chemotaxis simulation, it is significant to consider some of the optimal and extremely reactive behaviors of the heuristic approach under perfect conditions. In the heuristic approach to chemotaxis, perfect conditions relate to the tuning of gradient steepness and reward update hyperparameter such that the functions for changing velocity and direction at each step in agent simulation result in optimal moves in the gradient field. When the gradient is defined with such optimal conditions, the heuristic model is able to consistently and efficiently track through the gradient field, as illustrated in all results of optimal conditions of *Figure 5*. On top of this, as observed in countless cases in the result section of this paper, the heuristic model when leaving the central gradient of positive chemicals at optimal conditions will reverse direction, heading back into the gradient field, consistently trending towards the gradient center under optimal moves, regardless of position (*Figure 8*).

The abilities of the heuristic agent to perform optimally when provided perfect gradient information may at first not seem significant in the application of the model onto realistic chemotaxis, as it is impossible to rely on specific gradient information with the variability of real world microscopic chemical gradient fields. However, considering reality with a differing approach, the heuristic model can be specifically changed to react optimally to a given gradient field by adjusting key constants in the functions for reward update, changing direction, and velocity. If adjusted correctly with the knowledge of the gradient field which is being dealt with, it is a matter of simple experimentation and mathematical calculations in order to establish updated heuristic base models which can optimally travel through any gradient field – proving the possibility of more versatile and realistic chemotaxis for the model type. As heuristic models are specifically optimized to real world applications using the adjustments outlined in the prior section, the special abilities of heuristic models will come through in variations of conditions, showcasing the strengths of the model is simulating true chemotaxis.

4. Conclusions

The Q-learning model acting with consistency with more parallels to true chemotaxis opens up the possibility for confidence in model success even with uncertainties with the task at hand. Despite non specific chemical gradient information and somewhat unknown gradient location, Q-learning led microscopic robots and other small agents may still find success. If lack of knowledge is a critical issue in a task of microscopic navigation, the application of Q-learning methods can be pivotal in the success of a given task, as the methodology effectively finds success despite complications to gradient information. For Q-learning the task and gradient do not need to be as clearly outlined to bring success.

On the other hand, the heuristic model when optimized correctly provides a pipeline to optimal navigation, as the mathematical relationship required for efficient motion are established and can be benefited from. This can be effective when there is gradient information that is known to the point that such optimal relationships can be identified, leading microscopic robots and agents to effective and possible life saving navigation. As long as information is well known and recorded for a given task, the heuristic model can be turned to in a highly optimized (through various constants and changing parameters) form as a consistent and reactive solution to chemotactic like navigation that may bring relatively significant success. (For heuristic the task and gradient must be clearly outlined and understood).

Together the approaches of a reinforcement learning, Q-table based, model as well as the approach of an entirely rule based heuristic model, bring diverse strategies of microscopic navigation with the ability to approach diverse situations. Such versatile abilities should instill serious confidence in the abilities of humanity to interact with the microscopic world through modern advancements, leading miniature robots and manmade agents through the smallest of spaces. It is through these technological innovations in microscopic navigation strategies, which have been fleshed out in this paper, that humanity may be able to tackle some of the largest issues of the modern day, including various cellular diseases and cancerous issues, preventing death and bringing up quality of life on a broad level for the good of all.

Equations

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t) \right]$$

$$\sigma_f \leftarrow \sigma_i + m * \alpha \left[r_p + m * \gamma * r_p - \sigma_i \right], \; m = 3.75$$

$$v = \begin{cases} 4, & r_p{<}0 \\ \frac{1.5}{r_p}, & r_p{>}0 \end{cases} \qquad 1 < v < 4$$

$$\Delta\theta \leftarrow 180 \pm 180 \left(\frac{min(|\sigma_i|, |\sigma_f|)}{max(|\sigma_i|, |\sigma_f|)} \right)^m, \quad m = 0.6$$

$$\Delta\theta \leftarrow \pm 360 \left(\frac{\sigma_i}{\sigma_f} \right)^m, \quad m = 0.8$$

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